Diabetes Research and Clinical Practice, 7 (1989) 7–16
Elsevier

DRC 00286

Abnormal glucose tolerance in the Maltese

A population-based longitudinal study of the natural history of NIDDM and IGT in Malta

A.G. Schranz

Diabetes Clinic, St. Luke's Hospital, G'Mangia, Malta

(Received 4 April 1988)
(Revision received 11 October 1988)
(Accepted 17 January 1989)

Summary

A population-based longitudinal study of abnormal glucose tolerance in the adult Maltese, carried out within the WHO-assisted National Diabetes Programme, has recently been completed. During the 6-year interval abnormal as compared to normal glucose tolerance was found to be related to a significantly higher mortality: the age-adjusted relative risks of death were 3.3 times in diabetic females and > 2 times in IGT and diabetic males. In the repeat epidemiological survey 1422 subjects (66.8% of the initial sample) were reinvestigated with the oral GTT being interpreted according to WHO's 1985 recommendations. The age-standardised prevalence rates, in the 35–69-year-old males and females, were respectively 12.89% and 13.24% for IGT and 9.07% and 10.77% for diabetes. These gradually increased after age 40, IGT peaking in the 60 + year groups and diabetes 10 years later. Heredity (especially diabetes in close relatives) seemed a major influence, whilst excess body weight appeared the more important associated environmental factor. The incidence levels (% per annum) of diabetes during the interval were 0.71 for normoglycaemics and 5.1 for IGTs; this seven times higher risk in the latter was slightly lower in females than males, but significantly higher in the < 60-year-olds compared to older subjects. Of the initial IGTs 36% remained IGT and 33% reverted to normal glucose tolerance, whilst 11% of the initial normoglycaemics deteriorated to IGT. The determinants more strongly influencing worsening of glucose tolerance were age (> 50 years), baseline glycaemia (fasting > 5.5 mmol/l and a 2-h post-load glycaemia > 9.5 mmol/l) and initial body mass index (> 27 kg/m^2). In conclusion the data permit a better insight into the natural history of, and risk factors for, disturbed glucose tolerance in this community.

Key words: Glucose tolerance; Non-insulin-dependent diabetes mellitus; Malta

Background information

Among the major non-communicable diseases – the main causes of morbidity and mortality in the Maltese [10] – diabetes mellitus (DM), which has been known to occur in these small Mediterranean islands for at least the last 270 years, has for a long time appeared to play a significant role

Address for correspondence: Dr. A. G. Schranz, Diabetes Clinic, St. Luke's Hospital, G'Mangia, Malta.

0168-8277/89/$03.50 © 1989 Elsevier Science Publishers B.V. (Biomedical Division)

[1]. The fact that the estimated prevalence in an earlier study [2] of about 17% (albeit using 'soft' diagnostic criteria) of diabetes in Malta was particularly high, and that this disease with its complications was a considerable health problem in this semi-isolated community, led to the formulation of the National Diabetes Programme. Among its various specific aims were basic epidemiological investigations on the prevalence and incidence of, and characteristics related to, abnormal glucose tolerance in the adult population.

Study design and methodology

Within the framework of this Programme, a prospective follow-up on a cohort of Maltese people was performed in January 1987; this follow-up constitutes the subject of this paper. The main objective of phase V was basically a longitudinal study of the natural history of impaired glucose tolerance (IGT) and non-insulin-dependent diabetes mellitus (NIDDM) to investigate any changes that might have occurred during a 6-year interval in the Maltese.

The project consisted of a repeat epidemiological survey, on the same random representative sample of the local population as one carried out 6 years previously during phase I [3]. All persons now over 35 years of age who were investigated at that time were invited to participate, details being kept of deaths, emigrations, marriages and new addresses. The survey involved a standard questionnaire about personal medical history, family history of disease, and knowledge and behaviour about health, as well as a clinical examination (including anthropometry and blood pressure estimations) and an oral glucose tolerance test (GTT) on all subjects except known diabetics on treatment (tablets/insulin), the diagnosis and classification being done according to the WHO's 1985 recommendations.

Of the 950 male and 1178 female subjects who participated in phase I, 66.8% (624 men and 798 women, respectively 65.7% and 67.7%) were eventually re-investigated in phase V, thus rendering possible valid analysis of the various subgroups. Data were filled in appropriate forms and transferred for computer analysis which was done using mainly the SPSS software package.

Results

Prevalence of abnormal glucose tolerance

The levels of 13.1% IGT and 10% DM in the 35–69-year-olds detected in phase V amply confirmed the significant prevalence in the Maltese population of the abnormal glucose tolerance found in phase I. The age-standardised prevalence values (shown in Table 1) proved in this last phase to be 1.3 times higher in males for DM and more than twice as high in both sexes for IGT. The levels of newly diagnosed diabetics were four times higher in males (10.6% vs. 2.6%) and two times higher in females (7.9% vs. 3.7%) in the 45–74 age groups, whilst those of the previously known diabetics were similar. IGT was somewhat commoner in males whilst DM was more frequent in females. Both conditions de-

TABLE 1

Age-standardised prevalence (%) of IGT and diabetes in phases I and V by sex in age groups 35–69 of actual age

	IGT				Diabetes			
	Male		Female		Male		Female	
	n	%	*n*	%	*n*	%	*n*	%
Phase I	34	5.97	40	5.88	38	6.76	77	11.06
Phase V	61	12.89	78	13.24	45	9.07	65	10.77

teriorated with increasing age from 35 years onwards; however, whereas the former peaked in males and plateaued in females in the 60+ age groups, DM peaked in both sexes about 10 years later (Figs. 1 and 2).

No important differences between normal and diabetic subjects were seen in the socio-cultural factors of marital status, social class, income, education and occupation.

In both phases, those clinical symptoms usually associated with diabetes showed a tendency to increase in frequency as metabolic state deteriorated. They were two and 1.5 times (in females and males, respectively) more common in diabetics than in normals. The ones that could serve as the more useful diagnostic pointers for DM (as calculated from the relative risks (RR) in DM vs. normals) include asthenia (RR 3.1), polyphagia (RR 3.0), pruritus genitalis (RR 2.5), and polyuria (RR 2.1) in females; and unexplained weight loss (RR 3.5), polyuria (RR 2.1) and asthenia (RR 1.8) in males. Fasting urine samples containing > 0.5 mg/dl glucose were highly likely (RR 12.5) to show significant diabetes.

Heredity

A relevant proportion of subjects, particularly over age 50, gave a positive family history of diabetes which grew markedly stronger as the metabolic state of those interviewed deteriorated. The commonest findings were diabetes in maternal grandparents (circa 39%), in mother (circa 30%), in paternal grandparents (circa 28%), in

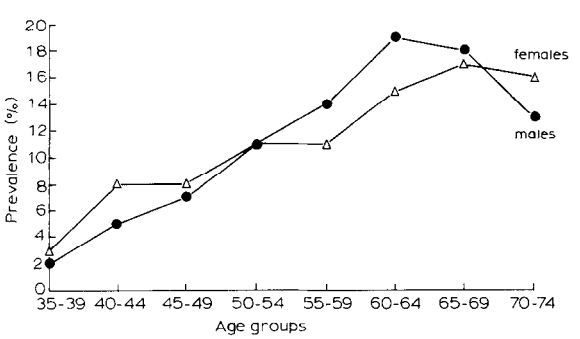

Fig. 1. Age distribution of IGT prevalence.

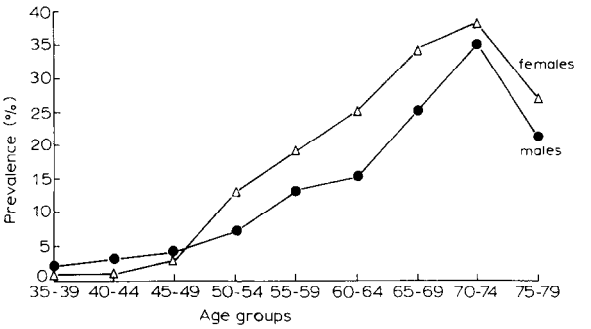

Fig. 2. Age distribution of diabetes prevalence.

father (circa 17%) and in siblings (circa 14%). The relative risk in diabetic compared to non-diabetic subjects of having a diabetic close relative are shown in Table 2, the highest being in siblings (esp. sisters) of females.

Environment

(a) Obesity. Excess body weight was found to be particularly common. Circa 42% of males and 31% of females were overweight (body mass index (BMI) 25–29.9) and another 20% of males and 34% of females were obese (BMI > 30). Adiposity started early in life (late 20s) in both sexes, overweight peaked at age 50 (men 58%, women 44%) and obesity 10 years later (men 36%, women 65%).

Mean BMI values (reflecting the body weight) increased with increasing age, being more pronounced in females than in males. On the other hand, the waist/hip ratios were as expected, persistently higher in males than in females in all age groups. However, whereas in men the mean ratio rose from 0.87 to a peak of 0.96 in the 40–49-year-olds, slowly dropping thereafter to 0.93, in women it kept rising gradually with age from 0.84 to 0.90. This increasing degree of android obesity in females was positively associated with the rising prevalence of diabetes seen in women as they grew older.

No differences of note were seen between normal IGT and DM cases in the mean BMI values, females having slightly higher levels in all categories, especially DM. However, statistically important increased prevalences of abnormal glu-

TABLE 2

Family history of diabetes in close relatives of subjects classified by metabolic status in phase V*

Relative	Subject							
	Male				Female			
	Normal		Diabetes		Normal		Diabetes	
	n	%	*n*	%	*n*	%	*n*	%
Mother								
DM yes	114	25.4	25	41.0	172	29.2	47	51.6
no	334	74.6	36	59.0	417	70.8	44	48.4
	$P = 0.05$		RR = 1.6		$P = 0.025$		RR = 1.8	
Father								
DM yes	73	16.5	16	29.1	86	15.3	16	17.9
no	370	83.5	39	70.9	477	84.7	73	82.1
	$P = 0.025$		RR = 1.8		P = NS		RR = 1.2	
Subjects	($n = 479$)		($n = 68$)		($n = 605$)		($n = 94$)	
Brothers	*n*	1069		177		1434		230
	% DM	7.8		24.3		7.3		21.3
	$P = 0.025$		RR = 3.1		$P = 0.025$		RR = 2.9	
Sisters	*n*	1140		183		1501		220
	% DM	6.5		21.9		5.9		23.2
	$P = 0.025$		RR = 3.4		$P = 0.025$		RR = 3.9	

* Cases where data were unknown or suspect were omitted.

TABLE 3

Metabolic status at follow-up by initial diagnosis

Metabolic status in phase V	Initial diagnosis in phase I						
	Total	Normal		IGT		Diabetes	
		n	%	*n*	%	*n*	%
Males							
Normal	484	472	84.4	11	31.4	0	0
IGT	74	60	10.7	12	34.3	2	5.9
Diabetes	71	27	4.9	12	34.3	32	94.1
Total	629	559	100	35	100	34	100
Females							
Normal	606	589	85.0	14	35.0	3	4.8
IGT	95	77	11.1	15	37.0	3	4.8
Diabetes	95	27	3.9	11	27.5	57	90.4
Total	796	693	100	40	100	63	100

TABLE 4

Age-adjusted means (and SD) of selected parameters in phase I of males and females classified according to their metabolic status in phase V

		Males			Females		
		Normal	IGT	DM	Normal	IGT	DM
Age (years)	mean	39.2	55.2	57.7	39.2	52.3	59.8*
	SD	15.8	10.2	12.0	14.2	12.3	9.1
Weight (kg)	mean	71.1	79.9	76.4	66.1	67.6	77.7
	SD	13.5	12.4	10.9	12.9	12.1	13.7
Glycaemia (mmol/l)							
Fasting	mean	4.1	4.7	6.2*	4.1	4.4	7.2*
	SD	0.7	1.3	2.6	0.7	1.0	3.1
Post-load	mean	4.6	6.1	9.7*	5.3	6.2	10.8*
	SD	1.3	2.2	4.7	1.5	2.3	5.5

* $P < 0.05$.

cose tolerance in both males and females were seen as the body weight increased. Compared to the lean, the prevalences of IGT and DM in the overweight were 1.6 and 3.1 times higher in males, and 2.0 and 3.3 times higher in females whilst in the obese they were 2.3 and 3.0 times in men and 2.3 and 4.8 times in women.

(b) Physical exercise. Both during work and free time no important differences were noted in the amount of exercise performed between the different groups of metabolic status.

(c) Energy intake. Although in both sexes (i) the estimated total energy intake was rather high (males averaging >3000 calories and females >2600) with a positive balance over energy expenditure of circa 10% in men and 19% in women, and (ii) the composition of macronutrient intake was unsatisfactorily balanced, no

TABLE 5

The risk of diabetes mellitus during a 6-year follow-up period in normoglycaemic and IGT subjects by sex

Diabetes at follow-up	Glucose tolerance at outset					
	Male		Female		All	
	Normal	IGT	Normal	IGT	Normal	IGT
Yes	27	12	27	11	54	23
No	532	23	666	29	1198	52
Total	559	35	693	40	1252	75
Cumulative 6-year incidence	4.8	34.3	3.9	27.5	4.3	30.7
IGT vs. normal						
RR (per sex)	7.15		7.05		7.14	
RR (normal at outset) = 0.81, RR (IGT at outset) = 0.80						

statistically significant differences were seen in the overall mean nutrient intake, even when expressed as % of total energy intake, between normal, IGT and DM cases.

Metabolic movement

During the 6-year follow-up period between phases I and V important changes occurred in the normal and IGT cases at baseline (Table 3). In the former 10.9% moved to IGT and 4.4% deteriorated to DM, whilst of the initial IGTs 36% remained IGT, 33.3% became normal and 30.7% worsened to DM. The cumulative 6-year incidence of DM was 4.3% in normals and seven times higher (30.7%) in IGTs.

Determinants of movement

The age-adjusted means of selected parameters in phase I of male and female subjects classified according to their metabolic status in phase V are shown in Table 4.

(i) Sex. No important differences were seen between sexes in relation to changes in metabolic status. The risk for the development of DM was only slightly lower in women (Table 5).

(ii) Age. This was predictive of abnormal glucose tolerance in both sexes, the means being higher in phase I in those found at follow-up to have IGT or DM rather than normoglycaemia but only statistically significant in female DMs (older by 19.8 years). Worsening of glucose tolerance occurred mainly in subjects over 60 years old, yet the higher risk for DM seen in IGTs compared to normals was more noted in younger persons (RR 7.18 vs. 3.26) (Table 6).

(iii) Family history of diabetes. Historically documented presence of diabetes in the parents of the subjects was associated with the deterioration of glucose tolerance mainly in those initially normoglycaemic. Statistically significant findings were only seen in cases of paternal diabetes related to deterioration to DM in males (RR 1.7) and to IGT in both sexes (RR 1.6), and of maternal diabetes related to worsening to DM in males (RR 1.7).

TABLE 6

The risk of diabetes mellitus during a 6-year follow-up period in normoglycaemic and IGT subjects by age

Diabetes at follow-up	Age at outset			
	Less than 60 years		60 years or more	
	Normal	IGT	Normal	IGT
Yes	37	9	17	14
No	1068	29	129	23
Total	1105	38	146	37
Cumulative 6-year incidence	3.3	23.7	11.6	37.8
IGT vs. normal				
RR (per age)	7.18		3.26	
RR (normal at outset) = 3.52, RR (IGT at outset) = 1.59				

(iv) Fasting blood glucose. The baseline levels of fasting glycaemia contributed significantly to the development of DM. The means were 2.1 mmol/l in men and 3.1 mmol/l in women higher in those found to have DM compared to normals at follow-up.

Whilst 20% of subjects with initial fasting blood glucose of 5.5–7.0 mmol/l had DM in phase V only 3% with levels <4.5 mmol/l at outset were later found to have DM. The influence of the baseline fasting glycaemic level on the incidence of DM was more pronounced in IGTs (RR men 3.1, women 1.6) (Table 7) and in people over age 60 (RR men 1.6, women 3.0), but the risk rose more sharply in normals and in younger subjects.

(v) 2-h Post-load glycaemia. This value in phase I was positively associated, in women slightly more than in men, with the development of DM especially in older subjects. In normals the risk was 3.2 (7% vs. 2.2%) times higher in those with levels > 5.0 mmol/l, whilst in IGT it was 2.8 (50%

TABLE 7

The risk of diabetes mellitus during a 6-year follow-up period in normoglycaemic and IGT subjects by baseline fasting glycaemia and sex

Initial fasting glycaemia (mmol/l)	Male				Female			
	Normal (n = 563) %	RR	IGT (n = 33) %	RR	Normal (n = 699) %	RR	IGT (n = 37) %	RR
<4.5	2.7	1.0	22.2	1.0	2.1	1.0	12.4	1.0
4.5–5.4	9.8	3.6	27.3	1.2	6.0	2.9	28.6	2.3
5.5–7.0	12.5	4.6	38.5	1.7	17.6	8.4	28.6	2.3

Relative risks: IGT vs. normal
Male <4.5 mmol/l = 8.2; 4.5–5.4 mmol/l = 2.8; >5.5 mmol/l = 3.1
Female <4.5 mmol/l = 5.9; 4.5–5.4 mmol/l = 4.8; >5.5 mmol/l = 1.6

vs. 17.9%) times greater in those with levels >9.5 mmol/l, compared to subjects with lower values of glycaemia.

(vi) Initial BMI. The age-adjusted mean BMI values in phase I in subjects found at follow-up to have IGT or DM were statistically significantly higher (11–17%) than the baseline means (men 26.2, women 28.1) of normoglycaemics, the differences being more pronounced in the 40–49 age groups. Similarly, the age-adjusted body weight in phase I seemed mainly predictive of IGT in males and DM in females, being on average 8.8 kg higher in the former and 11.6 kg higher in the

TABLE 8

Mortality between phases I and V in subjects per sex, age group and metabolic status at outset

	Normal		IGT		Diabetes	
	n	%	n	%	n	%
Males						
Total	480		51		57	
35–59	9	2.6	–	–	1	5.5
60–69	6	7.5	2	7.7	4	19.0
70+	13	25.0	11	64.7	10	55.5
Females						
Total	605		58		107	
35–59	7	1.5	–	–	4	10.2
60–69	5	6.3	–	–	6	15.4
70+	9	18.7	2	11.1	9	31.0
Age-adjusted RR of death			IGT vs. normal		DM vs. normal	
males			2.13*		2.37*	
females			1.00		3.31*	

* P = 0.025.

latter compared to the values of those normoglycaemic at follow-up.

(vii) Body weight and BMI changes. No significant difference was noted at the end of the 6-year interval in any of the mean BMI and weight values of subject groups classified according to their metabolic status at follow-up. The great majority of the minor changes occurred in the <40-year-olds.

Mortality

In the 6-year period a significantly higher relative number of deaths occurred in hyper- compared to normo-glycaemic subjects, particularly in those aged >60. The greatest risk was in DM (RR females 3.3, males 2.3) whilst in IGT the risk was only evident in males (RR 2.1) (Table 8).

In the case of diabetes, females also had more risk factors for coronary heart disease than males, especially obesity and hypertension, whilst in the case of IGT it was the other way round, with males having more hypertension, being more overweight and heavier smokers.

Discussion

Determinants of NIDDM

Malta has been confirmed to have a notably high prevalence of abnormal glucose tolerance. Compared to the populations with the highest prevalence rates of diabetes recorded in the literature (Pima Indians and Nauruans), the IGT levels in the Maltese are only mildly lower (13% vs. 16%), whilst the levels of diabetes are more clearly inferior (10% vs. 24%), yet still significant enough to place this community among those ethnic groups (Mexican Americans, Israelites, and migrant Asian/Fiji Indians) with the strongest susceptibility to NIDDM in the world [4,5]. The age distributions of the prevalence levels look interesting in that (a) the fall-off in the DM figures of both sexes, thought to be due to the survival of the fittest, seems to occur rather later than in the Nauruans and Pima Indians (70+ vs. 60+) and (b) there appears to be no notable fall-off in the figures of female IGTs — possibly in line with the lack of increased mortality seen in women.

The high prevalence of NIDDM here is reflected in the great disparity between type 2 and type 1 diabetes, in spite of the recent finding in Malta of an important prevalence of IDDM too [6]. The ratios between diabetes in adults 35–69 years old and in children 0–19 years old are 97 in males and 85 in females, being some nine times those seen in Europeans [7].

Data concerning family history of diabetes in Malta could be considered to be fairly reliable owing to the known high level of awareness about DM in our midst and to the often recollected specific therapy (tablets/insulin) taken by diabetic close relatives. It proved to be notably common, as might be expected from the high prevalence of DM here. The increased frequency of DM in female near relatives of diabetic subjects seems interesting. Together with the eight times higher incidence in Malta of gestational IGT compared to the local non-pregnant adult female population of the same age group [8], and the reported higher probability, in Pima Indians, of the later onset of obesity and DM in gestational diabetic mothers and their offspring [9], it raises intriguing possibilities on DM intervention.

Obesity seemed to be the more relevant environmental risk factor but since both it and diabetes are heterogeneous their interrelationship is not easy to interpret. Even though a higher BMI was associated with a stronger risk for glucose intolerance and, as in other studies, abdominal obesity with high waist/hip ratios in females [11,12] seemed related to diabetes, excess body weight on its own did not appear to offer a sufficient explanation of the high prevalence of diabetes in Malta. However, as recently suggested [22], the duration and perhaps the age at onset of adiposity may both be important determinants. Notwithstanding, the results of this study tend to agree with those in the Nauruans [4], highlighting the probable increased genetic susceptibility to DM in these people.

Physical inactivity and dietary mistakes (both

qualitative and quantitative) have been implicated [4,13] as possibly related to obesity and abnormal glucose tolerance, yet no strong association was found here between either of these two environmental risk factors and DM. However, females had a higher positive balance of energy, more obesity and a higher prevalence of diabetes than males. Moreover, there are recent indications [21] that between past phases high physical activity may have been more helpful in maintaining normoglycaemia than in improving hyperglycaemia.

On the other hand, the finding that a significant proportion of obese (circa 67%), physically less active (circa 33%), and/or 'big eaters' (among the 80% of the study population that stated to be on no weight-controlling diet) did not develop disturbed glucose tolerance could implicate the influence of other factors too, including the possibility of genetic or environmental protective factors.

Natural history of NIDDM

The high incidence levels of DM found in normoglycaemics (0.7% per annum) and IGTs (5.1% per annum) are in keeping with the high prevalence of diabetes in the Maltese. In the former these work out to be (like the prevalence figures) just less than half (0.4) the rate reported in the Nauruans [7] and are well in excess of incidence values in many other Caucasoid populations [5].

Similar to the situation in other countries [13,14], the risk for diabetes in Malta appeared to be greater in IGTs (particularly those with 2-h post-glucose-load levels of >9.5 mmol/l) than in normoglycaemics, whilst in accordance with reports in the Nauruans [18], about one third of the former remained IGT and another third (presumably the less severe) reverted to normoglycaemia. Also like in other communities [15], the principal predictors of future DM in the adult Maltese were age, baseline glycaemia (2-h post-load and fasting) and adiposity. Somewhat at variance with the findings in the Pima Indians [16] but like those from the U.S.A. [17], family history of diabetes (in the parents) was only a significant predictor of future diabetes in older normoglycaemic men, yet DM in the father appeared to be an important determinant for future IGT in normoglycaemics of either sex.

Although adiposity seemed to be an independent predictor of future abnormal glucose tolerance, it did not appear to be consistent in its effect. Diabetes incidence looked stronger than prevalence in the relationship with obesity, as in the case of the Nauruans [18], the baseline BMI value appearing the more important. The lack of any strong association between changes in weight and in glucose tolerance would seem to support the view that attempting to prevent rather than treat excess body weight could be more useful. Substantiating this would be the finding that the attainment and maintenance of a desirable body weight in the obese diabetic in Malta was neither easy to achieve nor particularly effective alone in time [19].

Mortality

The age-adjusted mortality rates from all causes in phase V were close to the ones recently found in Malta [10] and near to those reported in Fiji Indians who have similar DM prevalence rates [13]. As in other studies [23,24], the risk of death was highest in diabetes and intermediate in IGT. Whilst in the former it was more pronounced in females, in the latter it was, contrary to other reports [20], higher in males. The absence of excess mortality in female IGTs compared to normoglycaemic women could signify that in Malta IGT women, unlike NIDDMs, do not appear to lose much of their natural protection against cardiovascular disease, the major cause of death in DM here [10]. Further evaluation of the relative importance of abnormal glucose tolerance as a cause of death and of the contribution of associated risk factors (age, hypertension, smoking and obesity) so often concomitantly present seams justified.

Conclusions

The present study gives an interesting insight into the natural history and risk factors of abnormal glucose tolerance in Malta where this condition is strongly related to increased mortality, except possibly in female IGTs. The prevalence levels of 13% IGT and 10% diabetes in the 35–69-year-olds place this population among the ethnic groups with the highest levels of these disorders in the world. As in other communities with similar characteristics obesity seemed to be an important environmental risk factor and together with heredity appeared to have relevant but as yet insufficiently clear associated roles in aetiology, whilst age > 50 years, glycaemia levels in phase I (fasting > 5.5 mmol/l and 2-h post-load > 9.5 mmol/l),and baseline degree of adiposity (BMI > 27 kg/m^2) were the more significant determinants for the high rates of DM incidence seen in normoglycaemics (0.7% per annum) and IGTs (5.1% per annum). More research is evidently needed, but in the mean time these findings could allow the better delineation of the principal criteria for the selection in the Maltese of high-risk groups for diabetes, eventually leading to improved diabetes health care in the community.

References

1 Cassar, P. (1982) Historical developments of the concept of diabetes in Malta. Ministry of Health, Malta.

2 Zammit Maemple, J.V. (1965) Diabetes in Malta. Lancet ii, 1197.

3 Katona, G., Aganovic, I., Vuksan, V. and Skrabalo, Z. (1983) The National Diabetes Programme in Malta – Final Report Phases I & II. W.H.O. NCD/OND/DIAB/83.2, Geneva.

4 Zimmet, P. and King, H. (1986) The epidemiology of diabetes mellitus. In: K. G. M. M. Alberti and L. P. Krall (Eds.), The Diabetes Annual/2. Elsevier, Amsterdam, pp. 1–12.

5 WHO Study Group Report on Diabetes Mellitus (1985) WHO Technical Report Series 727, Geneva.

6 Schranz, A. G. and Prikazsky, V. (1988) IDDM in the Maltese islands. Submitted for publication.

7 Zimmet, P. and King, H. (1985) The epidemiology of diabetes mellitus: recent developments. In: K. G. M. M. Alberti and L. P. Krall (Eds.), The Diabetes Annual/1. Elsevier, Amsterdam, pp. 1–15.

8 Schranz, A.G. (1987) Gestational Diabetes in Malta. Ph. D. Thesis, Zagreb.

9 Pettit, D.J., Baird, H.R., Aleck, K.A. and Bennet, P.H. (1983) Excessive obesity in offspring of Pima Indian women with diabetes during pregnancy. N. Engl. J. Med. 308, 242.

10 Tuomilehto, J., Cacciottolo, J., Schranz, A.G. et al. (1989) Trends in mortality from major non-communicable diseases in the middle-aged population of Malta. Revue Epidemiol. Santé Publ. (in press).

11 Hautvast, J.G. and Duerenberg, P. (1987) The risks associated with obesity-epidemiological studies. In: Bender and Brooks (Eds.), Body Weight Control. Churchill Livingstone, Edinburgh, pp. 65–71.

12 Bjornstorp, P. (1987) The prevalence of obesity: complications related to the distribution of surplus fat. In: Bender and Brooks (Eds.), Body Weight Control. Churchill Livingstone, Edinburgh, pp. 72–80.

13 WHO Expert Committee on Diabetes Mellitus (1980) Second Report, Technical Series 646. WHO, Geneva.

14 Jarrett, R.J., Keen, H., Fuller, J.H. and McCartney, P. (1979) Worsening to diabetes in men with IGT. Diabetologia 16, 25.

15 Bennett, P.H., Knowler, W.C., Pettit, D.J. et al. (1982) Longitudinal studies of the development of diabetes in Pima Indians. In: Eschewege (Ed.), Advances in Diabetes Epidemiology. Elsevier, Amsterdam, pp. 65–74.

16 Knowler, W.C., Bennet, P.H., Pettit, P.J. and Savage, P.J. (1986) Diabetes incidence in Pima Indians: contributions of obesity and parental diabetes. Am. J. Epidemiol. 113, 144.

17 Butler, W., Ostrander, L.D., Carman, W.J. and Lamphier, D. (1982) Diabetes mellitus in Michigan. Prevalence, incidence and associated conditions. Am. J. Epidemiol. 116, 971.

18 King, H., Zimmet, P., Raper, L. and Balkau, B. (1984) The natural history of IGT in the Micronesian population of Nauru, a 6 year follow-up study. Diabetologia 26, 39.

19 Schranz, A.G. (1985) The Influence of Nutrient Intake on Obesity and Diabetes in Malta. M.Sc. Thesis, Zagreb.

20 Jarrett, R.J., McCartney, P. and Keen, H. (1982) The Bedford Survey: 10 year mortality rates. Diabetologia 22, 79.

21 Schranz, A.G., Tuomilehto, J., Marti, B., Jarrett, R.J. and Grabauskas, V. (1988) Low physical activity and worsening of glucose tolerance in Malta. Submitted for publication.

22 Jarrett, R.J. (1986) Descriptive Epidemiology of Diabetes in Diabetes Mellitus – Series in Clinical Epidemiology, Vol. 2. PSG, Boston, pp. 11–25.

23 Sicree, R., Ram, P., Zimmett, P. and King, H. (1985) Mortality and health service utilization amongst Melanesian and Indian diabetics in Fiji. Diabetes Res. Clin. Pract. 1, 227.

24 Zimmet, P., Finch, C., Schooneveldt, M. et al. (1988) Mortality from diabetes in Nauru. Diabetes Care 2, 305.